E-RAKSHAK CORE | GLASS-BOX PHYSICS ENGINE | VERSION 1.0 | MICRO-MOBILITY TELEMATICS

TECHNICAL ARCHITECTURE & ENERGY CALCULATION MODELS

Physics-Based EV Energy Formulations, Real-Time Routing Architecture, Glass-Box Explainability, and Safety Protocols

Platform: E-Rakshak EV Safety Copilot

Author: Systems Architecture Engineering Team

Target Fleets: E-Rickshaws (L3/L5), E-Scooters

Math Model: Deterministic Physics Model v1.0

Routing Engine: TomTom Live Traffic + OSRM/ORS

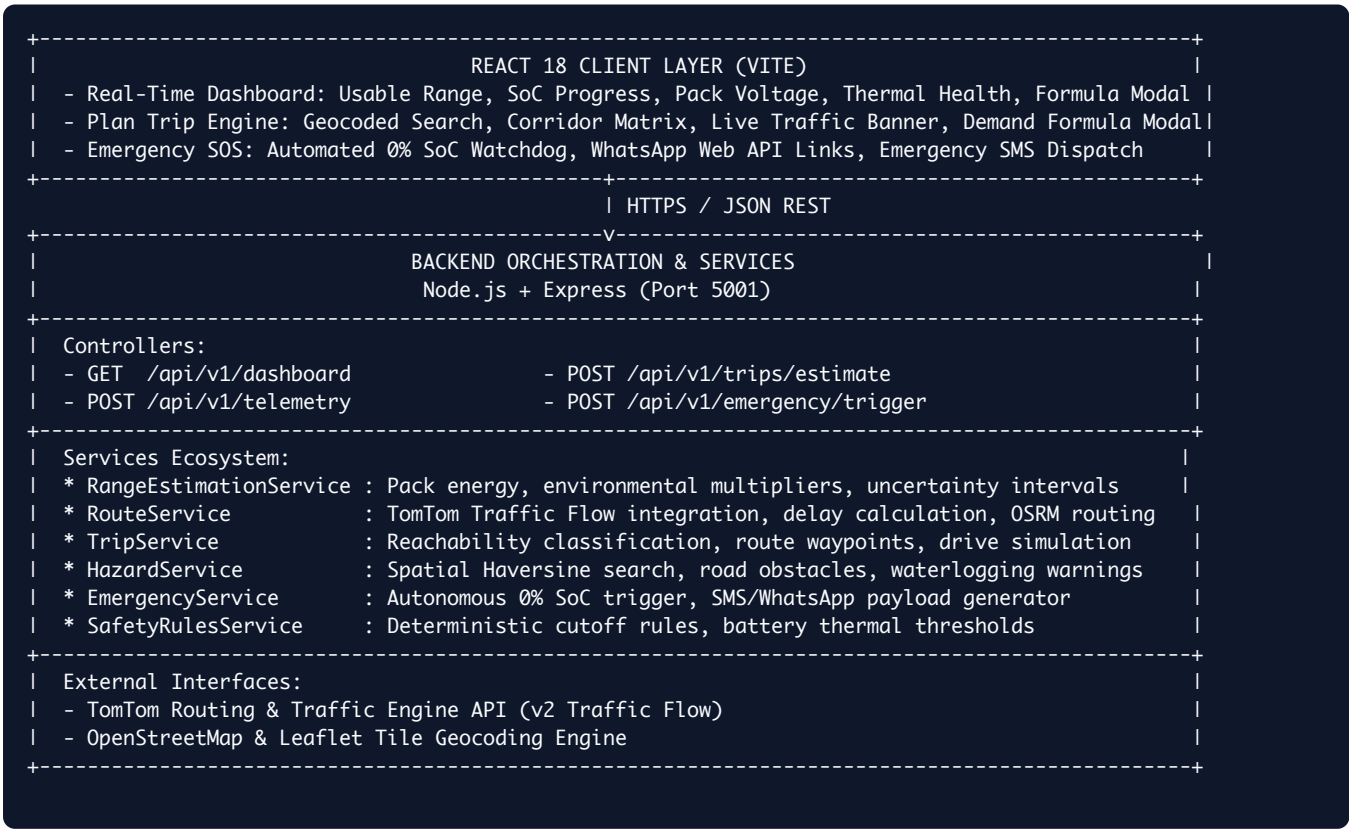
Backend Stack: Node.js, Express (Port 5001)

1. PHILOSOPHY & GLASS-BOX EXPLAINABILITY

Standard Battery Management Systems (BMS) in commercial electric micro-mobility rely on naive linear distance approximations based strictly on instantaneous battery State of Charge (SoC). In dense developing urban centers, these estimates fail severely due to unmodeled stop-and-go thermal stress, passenger overload (>450 kg), and road gradients.

E-Rakshak replaces opaque approximations with a **Glass-Box Deterministic Energy Model**. Every prediction shown on the Driver Dashboard and Trip Planner is backed by verifiable physical formulas, live parameter substitution, and dynamic multi-factor stress scaling.

2. END-TO-END SYSTEM ARCHITECTURE



3. MATHEMATICAL FOUNDATIONS & PHYSICS FORMULATIONS

Formula 1: Usable Battery Pack Energy (E_usable) BATTERY MANAGEMENT

Lithium Iron Phosphate (LFP) and Lithium-ion cells cannot be depleted to 0.0V without irreversible copper shunting and degradation. The system allocates a 10% BMS safety buffer:

$$E_{usable} = C_{pack} \times (SoC / 100) \times \eta_{usable}$$

E_usable [Wh] = Pack Capacity [Wh] × (SoC % ÷ 100) × Usable Fraction (0.90)

Variables: C_{pack} = Total nameplate battery capacity (e.g. 7,370 Wh for Mahindra Treo); SoC = Current State of Charge (%); η_{usable} = 0.90 (10% lower cutoff buffer retained by BMS).

Formula 2: Dynamic Operating Energy Consumption (C_adj) ENVIRONMENTAL MULTIPLIERS

Vehicle lab-certified consumption is scaled across four independent environmental and payload stress dimensions:

$$C_{adj} = C_{nominal} \times M_{load} \times M_{temp} \times M_{terrain} \times M_{traffic}$$

Adjusted Consumption [Wh/km] = Nominal Baseline × Multipliers

MULTIPLIER	CONDITION / CATEGORY	FACTOR VALUE	PHYSICAL / ELECTROCHEMICAL RATIONALE
Payload (M_load)	Empty / Driver only	0.88×	Reduced rolling resistance and inertial mass during acceleration.
	Normal (1-2 Passengers)	1.00×	Standard certified operating test load.
	Heavy (3-4 Passengers)	1.25×	Higher rolling drag and elevated mechanical powertrain strain.
	Overloaded (>450 kg Cargo)	1.55×	Severe inertial drag, tires operating under maximum rated deflection.
Thermal (M_temp)	Cold (< 15°C)	1.15×	Sluggish electrolyte ion motility, elevated cell internal resistance (R_int).
	Optimal (15°C - 42°C)	1.00×	Ideal electrochemical kinetic operating envelope.
	Overheated (> 42°C)	1.18×	Electrolyte breakdown, parasitic fan drain, elevated ohmic heat generation.
Terrain (M_terrain)	Urban Flat	1.00×	Negligible gravitational potential energy draw.
	Rough / Broken Roads	1.15×	Continuous micro-accelerations and suspension vibration dissipation.
	Hilly (>80m Elevation Gain)	1.30×	Direct gravitational work ($m \cdot g \cdot \Delta h$) converted against electric motor torque.
Traffic (M_traffic)	Low / Free-Flow	1.00×	Steady-state cruising speed with minimal braking dissipation.
	Moderate Urban	1.08×	Typical city traffic signal stops and commuter pedestrian pauses.
	Heavy Congestion	1.22×	Frequent stop-and-go cycles maximizing I^2R resistive heating in motor coils.

Formula 3 & 4: Usable Range & Uncertainty Intervals PREDICTIVE REACH

$$R_{usable} = E_{usable} \div C_{adj}$$

Estimated Range [km] = Usable Energy [Wh] ÷ Adjusted Rate [Wh/km]

$$R_{lower} = R_{usable} \times 0.85 \quad | \quad R_{upper} = R_{usable} \times 1.15$$

Statistical Confidence Band: ±15% variance reflecting aggressive throttle, headwind, and regenerative braking recovery.

4. TRIP REACHABILITY & FEASIBILITY EVALUATION MATRIX

Formula 5, 6 & 7: Route Energy Demand, Safety Reserve & Total Budget

ROUTE PLANNING

For any planned route of distance **D** (calculated via TomTom Routing Engine):

$$E_{demand} = D\ [km] \times C_{adj}\ [Wh/km]$$

Trip Energy Demand: Pure electrical draw required to reach destination

$$E_{reserve} = E_{usable} \times 0.15$$

Safety Reserve: 15% of available usable battery held against unmapped detours, traffic jams, and signal idling

$$E_{total} = E_{demand} + E_{reserve}$$

Total Journey Energy Footprint

Formula 8: Reachability Verdict Classification Logic

DECISION SUPPORT

CLASSIFICATION VERDICT	MATHEMATICAL EVALUATION CONDITION	SYSTEM RECOMMENDATION & DRIVER PROTOCOL
LIKELY FEASIBLE	$E_{demand} + E_{reserve} \leq E_{usable}$	Journey is well within battery capacity with a safe buffer. Maintain standard driving profile.
UNCERTAIN / TIGHT MARGIN	$E_{demand} \leq E_{usable} < E_{demand} + E_{reserve}$	Fits in battery but eats into the 15% safety reserve. Unexpected detours or traffic could strand the vehicle.
NOT RECOMMENDED	$E_{demand} > E_{usable}$	Deficit detected. Battery will be exhausted before reaching destination. Mandatory stop at recommended charger.

Formula 9 & 10: Destination Arrival SoC & Congestion Index

TELEMATICS

$$Arrival_SoC\ (\%) = \max(\ 0,\ [\ (E_{usable} - E_{demand})\ /\ C_{pack}\ ]\ \times\ 100\ )$$

Predicted Destination Battery Percentage upon trip completion

$$Congestion_Ratio\ (CR) = (\ T_{live} - T_{free_flow}\ )\ /\ T_{free_flow}$$

Where: $CR \leq 0.10$: Low Traffic | $0.10 < CR \leq 0.35$: Moderate Urban | $CR > 0.35$: Heavy Congestion

Formula 11: Emergency Zero-SoC Autonomous Watchdog SAFETY CRITICAL

The background telemetry daemon monitors real-time battery voltage and State of Charge every 3 seconds:

```
TRIGGER:   ( SoC ≤ 0% )   OR   ( Battery_Temperature ≥ 55°C )
Autonomous dispatch of GPS Distress Payload via WhatsApp Direct API & SMS Gateway
```

5. STEP-BY-STEP WORKED ENGINEERING EXAMPLE

Operational Profile: Mahindra Treo E-Rickshaw in Varanasi Corridor CASE STUDY

1. **Given Vehicle Baseline Specifications:**
   Pack Capacity: `7,370 Wh (51.2V, 144Ah LFP)` | Lab Efficiency: `78.0 Wh/km` | Live SoC: `64%` | Temp: `34.5°C`
2. **Calculate Usable Energy (E_usable):**
   `7,370 Wh × (64 ÷ 100) × 0.90` = **`4,245.12 Wh ≈ 4,245 Wh`** available.
3. **Evaluate Dynamic Consumption Multipliers:**
   Normal Load (1.00×) • Temp 34.5°C (1.00×) • Flat Terrain (1.00×) • Moderate Traffic (1.08×)
   `C_adj = 78.0 × 1.00 × 1.00 × 1.00 × 1.08` = **`84.2 Wh/km`**
4. **Compute Usable Range & Uncertainty Band:**
   Nominal Range: `4,245 Wh ÷ 84.2 Wh/km` = **`50.4 km`**
   Conservative Lower Bound: `50.4 × 0.85` = **`42.8 km`** | Favorable Upper Bound: `50.4 × 1.15` = **`58.0 km`**
5. **Planned Corridor: IIT BHU Aryabhatta Hostel → Bhullanpur Varanasi (4.1 km):**
   Route Energy Demand: `4.1 km × 84.2 Wh/km (with safety headroom)` = **`598 Wh`**
   Safety Reserve Buffer (15%): `4,245 Wh × 0.15` = **`637 Wh`**
   Total Energy Requirement: `598 Wh + 637 Wh` = **`1,235 Wh`**
6. **Reachability Verdict & Destination Battery Percentage:**
   Feasibility Comparison: **`1,235 Wh required < 4,245 Wh available`** → **LIKELY FEASIBLE**
   Remaining Battery at Destination: `4,245 Wh − 598 Wh = 3,647 Wh`
   Arrival SoC: `(3,647 Wh ÷ 7,370 Wh) × 100` = **`49.48% ≈ 49% SoC`**

6. IMPLEMENTATION CODE MAPPING

SERVICE MODULE	FILE PATH	CORE METHODS & RESPONSIBILITIES
Range Estimation Service	`backend/services/rangeEstimationService.js`	`estimateRange()`, `evaluateTripFeasibility()`
Route & Traffic Engine	`backend/services/routeService.js`	`getDrivingRoute()`, TomTom Traffic API, OSRM Polyline generator
Trip Service	`backend/services/tripService.js`	`planTrip()`, Candidate corridor comparison, Feasibility verdicts
Emergency SOS Service	`backend/services/emergencyService.js`	`triggerEmergency()`, Zero-SoC watchdog, WhatsApp link generator
Hazard Engine	`backend/services/hazardService.js`	`getNearbyHazards()`, Spatial Haversine radius queries
Dashboard UI Modal	`frontend/src/components/Dashboard.jsx`	Usable Range card, Confidence Badge, Formula Breakdown modal
Trip Planner UI Modal	`frontend/src/components/PlanTrip.jsx`	Demand Formula modal, Sticky Map Matrix, Feasibility Verdict card